
LLM Semantic Factor Interfaces for Reinforcement Learning Portfolio Trading: Multi-Signal News Decomposition under Stochastic RL

Abstract

Many sequential decision systems occasionally observe *sparse* unstructured text: informative language arrives on a minority of steps, while most transitions carry only a neutral default. **This paper** empirically studies how such text should enter *portfolio* decision state—as dense features, a single scalar, or a small set of interpretable coordinates—and how to separate representation hypotheses from optimisation variance when a frozen LLM supplies the encoding. We instantiate **Semantic State Abstraction Interfaces** (SSAI): prompted maps from documents to K auditable axes with neutral defaults on no-news days (Definition 1, Appendix), here with $K = 4$ (*sentiment, risk, confidence, volatility forecast*) on a **FinRL-MultiSignal** US-equity panel (30 NASDAQ-100 names; FNSPID news; train 2013–2018; test 2019–2023). Direct portfolios (SFP/SRF/SCW), supervised ridge forecasts, and paired RL agents (DP-PPO, SAC) share the same ϕ so signal and optimiser effects can be read off separately. On the test window, four-factor SFP reaches **307.2%** cumulative return and Sharpe **1.067** vs. **291.9%/1.050** sentiment-only and **243.6%/1.032** equal-weight buy-and-hold; residualizing non-sentiment axes preserves gains; conviction weighting reaches **314.1%**. Naively appending sparse LLM columns hurts a dense price/technical ridge forecaster, while lexical headline aggregates (VADER; TF-IDF/SVD) and a validation-selected high-conviction semantic tilt recover Sharpe near the price-only ridge frontier—consistent with SSAI as a *sparse allocation interface* rather than a naive dense LLM block. The RL block is *diagnostic*, not a performance claim: DP-PPO trails buy-and-hold; SAC with identical SSAI observations nearly matches buy-and-hold return and improves Sharpe (**1.128** vs. **1.032**), isolating algorithm dependence. Across 21 DP-PPO seeds, full scoring raises mean return ($195.0\% \pm 44.2\%$ vs. $190.9\% \pm 42.6\%$) and Sharpe (0.920 ± 0.096 vs. 0.907 ± 0.094) versus neutral-masked *evaluation* of the same checkpoints, but paired Wilcoxon tests remain non-significant ($p \approx 0.25$ – 0.31), and bootstrap 95% intervals for seed-mean Sharpe overlap substantially—so masking evidence is descriptive, not confirmatory at this sample size. Overall, the strongest evidence for multi-axis LLM semantics is in direct portfolios and supervised sparse-use designs; RL results are stress tests under heavy neutral-default regimes, with cross-domain SSAI claims left as a reusable template rather than an additional empirical claim of this submission.

1 Introduction

Deep Reinforcement Learning (DRL) is widely studied for sequential decision problems where observations mix numerical signals with unstructured evidence [Mnih et al., 2015, Schulman et al., 2017, Liu et al., 2021]. Large language models (LLMs) offer flexible encoders for text [Brown et al., 2020, Liu et al., 2023, Wu et al., 2023], yet most pipelines obscure a basic design question: should textual evidence appear as a high-dimensional latent vector, as a single scalar (e.g., sentiment), or as a small inventory of named semantic coordinates—especially when text arrives only on a sparse subset of decision dates?

Prior work has connected NLP with RL trading in limited ways. Yang et al. [2020a] use technical indicators as the sole state representation; Benhenda [2025] add a single sentiment score from DeepSeek-V3; and Lopez-Lira and Tang [2023] use ChatGPT to classify daily market sentiment as a binary signal. A single sentiment score is information-sparse: it conflates orthogonal concepts—whether a headline is positive differs fundamentally from whether the underlying situation is *risky* or whether the market is likely to be *volatile*.

Positioning. We adopt *Semantic State Abstraction Interfaces* (SSAIs), formally defined in Appendix A: maps ϕ from document sets $\mathcal{N}_{s,d}$ to K interpretable coordinates elicited by fixed prompts from a frozen LLM, with a neutral default when $\mathcal{N}_{s,d} = \emptyset$. Methodologically, the contribution is not a new RL gradient formula but (i) a reusable SSAI template—auditable axes, sparse-by-design neutral fills, shared ϕ across estimators—and (ii) an evaluation protocol that separates representation hypotheses from optimisation variance: direct factor portfolios and supervised dense-vs-sparse regressions read off semantic signal with minimal policy entropy; DP-PPO and SAC supply paired nonlinear stress tests on identical SSAI observations. **Scope:** Definition 1 is domain-agnostic, but **all empirical validation in the main text** is on the US equity panel above; Appendix D lists speculative cross-domain instantiations without experiments.

Trading serves as a demanding instantiation because aggregate outcomes are noisy, textual coverage is uneven (14.8% non-neutral stock-days), and dense ridge augmentation provides an explicit “wrong” use of sparse semantics against which SSAI tilts and factor portfolios contrast.

Research questions. We organise the experiments around three concrete questions: **RQ1:** do multi-axis LLM signals change policy behaviour relative to neutral-masked inputs? **RQ2:** are any LLM axes aligned, even weakly, with realised market quantities? **RQ3:** is observed performance primarily a property of the semantic state representation, or of the learning algorithm used to exploit it? The results answer these conservatively: direct semantic portfolios favour the four-factor representation over sentiment-only, SRF preserves that result after removing linear sentiment components from the remaining axes, SCW improves allocation use of semantic score magnitudes, supervised price forecasting benefits only when semantic scores are used as a validation-selected high-conviction tilt rather than naively appended as dense columns, DP-PPO signal effects are directional but *statistically underpowered* at 21 seeds (Wilcoxon non-significant; overlapping bootstrap CIs), ICs are small but interpretable, and SAC substantially improves risk-adjusted outcomes under the same LLM state.

We make three contributions:

1. **Semantic state abstraction (SSAI template).** We instantiate Definition 1 with $K=4$ interpretable 1–5 axes—sentiment, risk, confidence, volatility forecast—aggregated to ticker-day states with neutral defaults and shared across portfolios, supervised learners, and RL agents.
2. **Isolation tests for semantic factorization.** We introduce SFP, SRF, and SCW to test the four-axis representation outside RL variance: SFP learns linear semantic weights, SRF removes linear sentiment components from non-sentiment axes, and SCW uses validation-selected score magnitude for portfolio concentration. A supervised price/technical stress test further separates useful sparse semantic tilts from naive dense feature appending.
3. **Representation-versus-optimization diagnostics (RL as stress test).** We evaluate the same SSAI observations under DP-PPO and SAC with masking, sub-period, transaction-cost, and multi-seed analyses. The RL headline is *algorithm dependence*, not beat-the-market claims: DP-PPO illustrates fragile exploitation under sparse signals, whereas SAC nearly closes the passive benchmark gap on returns while improving Sharpe—evidence that identical SSAI states interact very differently with optimiser dynamics.

2 Related Work

Liu et al. [2021] introduce FinRL with PPO and SAC baselines [Yang et al., 2020b, Liu et al., 2022]; risk-constrained RL has used CVaR objectives [Tamar et al., 2015] and Lagrangian relaxation [Ray et al., 2019]. FinGPT [Liu et al., 2023] and BloombergGPT [Wu et al., 2023] fine-tune LLMs on financial corpora; Benhenda [2025] add a scalar DeepSeek-V3 sentiment score to RL state. We extend prior scalar sentiment interfaces [Benhenda, 2025, Lopez-Lira and Tang, 2023] with SSAIs

instantiated through zero-shot multi-axis prompting (no encoder fine-tuning) and attach diagnostic evaluations across portfolios, supervised learners, and RL. News data are from FNSPID [Dong et al., 2024]. **Baseline scope.** We isolate SSAI design choices with deterministic portfolios and matched supervised learners; lexical headline aggregates (VADER; train-fit TF-IDF/SVD) now appear as dense ridge inputs (Table 3). We still do *not* report full FinBERT/FinGPT encoder pipelines or Transformer price-news forecasters matched on leakage and compute—those remain the strongest neural encoder comparisons for future work (Discussion).

3 Method

Schematic of the news-to-policy pipeline is Figure 2 (Appendix B).

3.1 Multi-Signal LLM Scoring

Given a news article headline or summary t about stock ticker s , we issue a single structured prompt to an LLM and extract four integer scores $\sigma = (\sigma_{\text{sent}}, \sigma_{\text{risk}}, \sigma_{\text{conf}}, \sigma_{\text{vol}}) \in \{1, \dots, 5\}^4$.

σ_{sent} **Sentiment:** 1 = very negative, 5 = very positive.

σ_{risk} **Risk:** 1 = very low company/market risk, 5 = very high risk.

σ_{conf} **Confidence:** 1 = very uncertain outlook, 5 = very certain.

σ_{vol} **Volatility forecast:** 1 = very calm expected price, 5 = highly volatile.

We use explicit axes rather than dense article embeddings because the study is designed to audit a state interface, not merely maximize predictive capacity. The integer coordinates support ticker-day coverage checks, residualization against sentiment, leave-one-axis masking, and deployment-facing interpretation of why a policy or ranking rule changes exposure. Dense latent text encoders and price-news Transformers are natural stronger baselines, but they do not provide the same axis-level causal and diagnostic handles without additional attribution machinery.

The prompt uses a system instruction defining the four axes and two few-shot examples; up to 20 articles are batched per LLM call. We apply this protocol to 40,850 articles from the FNSPID NASDAQ subset, scoring 39,995 (97.9%) successfully. Score summaries appear in Table 10 (Appendix F); risk and volatility are right-skewed (mean ≈ 2.5), consistent with adverse-news skew in financial headlines.

Signal coverage is uneven after aggregation to the trading panel: This sparsity is a central reason we treat the interface as a weak semantic factor rather than a dense predictive signal. Importantly, the direct portfolio experiments already control for this: SFP uses signal deviations only on days with non-neutral coverage, and outperforms equal-weight buy-and-hold by 63.6pp over 2019–2023—a gap that cannot be attributed to coverage volume alone since mean full-scoring evaluation exceeds neutral-masked evaluation of the same checkpoints across 21 seeds (Table 6).

Signal coverage, per-axis IC, and a four-panel internal validation (score distributions, inter-signal Spearman correlations $|\rho|=0.58\text{--}0.81$, lag-1 autocorrelation 0.47–0.57, and confidence IC $p=0.004$) are reported in Appendix F (Figure 4, Tables 11, 12). The key takeaway: scores are not white noise—they exhibit the directional cross-axis structure expected from financial news and are temporally persistent at the ticker level.

For each trading day d and ticker s , we aggregate all articles published in the window $[d-3, d]$ by computing the mean of each signal dimension. Days with no matching articles retain a neutral value of 3.0.

3.2 Trading Environment

We build on the FinRL StockTradingEnv gymnasium environment. The state vector on day d is:

$$\mathbf{s}_d = [c_d, \mathbf{p}_d, \mathbf{h}_d, \mathbf{f}_d, \boldsymbol{\sigma}_d] \in \mathbb{R}^{1+4N+KN} \quad (1)$$

where c_d is available cash, $\mathbf{p}_d \in \mathbb{R}^N$ are closing prices, $\mathbf{h}_d \in \mathbb{R}^N$ are current share holdings, $\mathbf{f}_d \in \mathbb{R}^{KN}$ are $K = 7$ technical indicators (MACD, Bollinger upper/lower, RSI₃₀, CCI₃₀, ADX₃₀,

138 30-day SMA, 60-day SMA) for each of $N = 30$ stocks, and $\sigma_d \in \mathbb{R}^{4N}$ are the four LLM signals.
 139 The total dimension is $1 + 2 \times 30 + (7 + 4) \times 30 = 421$.

140 Actions $a_d \in [-1, 1]^N$ are scaled by a maximum trade size $h_{\max} = 100$ shares. Transaction costs of
 141 0.1% per trade are applied. A turbulence index [Kritzman and Li, 2010] triggers a no-buy constraint
 142 when it exceeds a threshold of 380, implementing a simple market-regime filter.

143 3.3 Drawdown-Penalised PPO Reward

144 The per-step reward applies *drawdown shaping*:

$$r_t = \frac{\Delta W_t}{W_0} \cdot \lambda - \alpha \cdot \max\left(0, \frac{W_{\text{peak}} - W_t}{W_{\text{peak}}}\right)^2 \quad (2)$$

145 where $\Delta W_t = W_t - W_{t-1}$ is the change in portfolio value, $\lambda = 10^{-4}$ is the reward scale, W_{peak}
 146 is the running maximum portfolio value, and $\alpha = 0.1$ is the drawdown penalty coefficient. The
 147 quadratic form penalises large drawdowns super-linearly, encouraging the agent to protect gains
 148 rather than risk them for marginal upside (Appendix B.6 interprets the shaping term).

149 We train with a multi-process PPO implementation (OpenAI SpinningUp), using 4 MPI workers,
 150 hidden layers (512, 512) with Tanh activations, learning rate 3×10^{-4} , clip ratio 0.2, and 30 epochs
 151 over the 2013–2018 training set ($\approx 45,300$ daily observations across 30 stocks). **Naming.** Released
 152 checkpoints use the CPP0 filename prefix for historical compatibility; all tables label this agent
 153 **DP-PPO** (drawdown-shaped reward, Sec. 3.3), not Lagrangian constrained optimisation.

154 3.4 LLM Signal Integration

155 All four LLM channels enter σ_d as described above. We do *not* encode hand-crafted trading rules
 156 from individual scores; any dependence of actions on sentiment, risk, confidence, or volatility views
 157 is learned end-to-end by the policy network.

158 **Semantic Factor Portfolio (SFP).** To test whether the four LLM axes contain usable structure
 159 independent of RL, we also define a non-RL semantic factor portfolio. On the 2013–2018 training
 160 panel, SFP fits a ridge-regularised linear model from signal deviations ($\sigma - 3$) to 5-day forward
 161 returns. At test time, it scores each stock-day using the learned semantic factor weights, forms a daily
 162 long-only top-10 portfolio, and applies the same 0.1% transaction-cost assumption. The ridge penalty
 163 is fixed at 10^{-3} for all SFP variants rather than tuned on the test set. We report both the full four-factor
 164 SFP and a sentiment-only SFP, making the decomposition advantage testable without policy-gradient
 165 stochasticity. To control for the possibility that all additional axes merely recode sentiment, we also
 166 define *Semantic Residual Factorization* (SRF). SRF keeps sentiment as the anchor feature and, on the
 167 training panel only, fits $\sigma_j = a_j + b_j \sigma_{\text{sent}} + \epsilon_j$ for $j \in \{\text{risk, confidence, volatility}\}$. The portfolio
 168 scorer is then fit on $(\sigma_{\text{sent}}, \epsilon_{\text{risk}}, \epsilon_{\text{confidence}}, \epsilon_{\text{volatility}})$, testing whether residual semantic axes add
 169 structure beyond scalar sentiment under the same linear ranking rule. We also introduce *Semantic*
 170 *Conviction Weighting* (SCW). Instead of assigning equal weights to the top-10 semantic names, SCW
 171 applies a softmax over their learned factor scores and allocates more capital to higher-conviction
 172 names. The softmax temperature is selected on the final training year (2018) by validation Sharpe
 173 and then fixed for the 2019–2023 test window, avoiding test-period tuning.

174 4 Experimental Setup

175 **Data.** We use Yahoo Finance OHLCV for 30 liquid NASDAQ-100 constituents (tickers listed in
 176 Appendix B.4). Training: 2013-01-02 to 2018-12-31 ($\approx 45,300$ observations). Out-of-sample testing:
 177 2019-01-02 to 2023-12-29 (1,258 trading days).

178 News articles are sourced from the FNSPID dataset [Dong et al., 2024] via Hugging Face
 179 (Zihan1004/FNSPID), filtered to the same 30 tickers. 40,850 articles are collected; 39,995 (97.9%)
 180 are successfully scored by the LLM via a hosted chat-completions API.

181 **Baselines.** We compare against:

- **Semantic Factor Portfolio variants (SFP/SRF/SCW)**: non-RL long-only top-10 portfolios using ridge-learned weights over all four LLM axes, sentiment alone, SRF residual axes after controlling for sentiment, or SCW conviction weighting inside the selected basket (Table 2).
- **Supervised ridge forecasters**: top-10 portfolios ranked by 5-day return forecasts from price/technical features alone, price plus LLM sentiment or four-factor semantics, price plus lexical headline dense features (VADER; train-fit TF-IDF/SVD on the same headline window), or a price forecast with a validation-selected high-conviction semantic tilt (Table 3; optional FinBERT tone via `dense_text_baselines.py -finbert`).
- **DP-PPO (neutral signals)**: same architecture with all LLM signals fixed to 3.0 at evaluation time, isolating textual variation.
- **SAC (LLM signals)**: an off-policy Soft Actor-Critic baseline with the same four-signal observation vector, isolating algorithm choice from textual-feature design (Table 4).
- **Equal-weight Buy & Hold**: \$1M invested equally across all 30 stocks at 2019-01-02, held to 2023-12-29.
- **Momentum (top-10) and Equal-Vol (risk parity)**: rule-based baselines from the same price panel (see Table 1).

Metrics. Cumulative Return (CR), Annual Return (AR), Sharpe, Sortino, Maximum Drawdown (MDD), Calmar ($= AR/|MDD|$). Transformer and constrained-risk baselines remain future work (see Section 2).

5 Results

5.1 Main Performance Comparison

Table 1 reports all metrics on the 2019–2023 test period (numeric cells match `eval_harness.py` exports).

Table 1: Main Strategy Comparison (2019–2023)

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Rachev	CVaR-5%	Calmar
DP-PPO (all signals)	190.431	0.917	1.216	-42.540	0.911	-4.169	0.560
Momentum (top-10)	117.566	0.664	0.922	-51.027	0.971	-4.406	0.330
Equal-Vol (risk parity)	230.661	0.989	1.315	-38.523	0.978	-4.110	0.703
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.969	-4.008	0.764

Key findings. (1) **Decomposition beats sentiment-only.** SFP reaches **307.2%** CR / Sharpe **1.067** vs. **291.9%/1.050** (sentiment-only) and **243.6%/1.032** (buy-and-hold); SRF preserves this after residualizing non-sentiment axes; SCW improves further to **314.1%** using score magnitude for conviction weighting (Table 2). (2) **Lexical dense text vs sparse LLM semantics under ridge.** Price-only ridge: **126.4%/0.653**; naively appending all four LLM semantic columns hurts; headline aggregates aligned with the LLM news window help when appended as dense columns—VADER: **134.8%/0.674**; TF-IDF/SVD: **132.8%/0.673**—while high-conviction semantic tilt matches this Sharpe regime (**133.9%/0.674**; Table 3). (3) **Algorithm dominates representation.** SAC with the same four LLM signals: **242.7%/Sharpe 1.128**, nearly matching buy-and-hold; DP-PPO trails by 53pp CR. Mechanism: SAC’s replay buffer revisits the rare semantic-rich transitions that on-policy PPO dilutes (Table 4). (4) **Multi-seed (descriptive).** Eight seeds: CR **193.7% $\pm$ 43.7%**, Sharpe **0.913 $\pm$ 0.094** (full) vs. 187.2% $\pm$ 42.9%, 0.889 $\pm$ 0.084 (neutral-masked); bootstrap 95% CIs for seed-mean Sharpe overlap substantially ([0.85, 0.98] vs. [0.83, 0.94]); Wilcoxon $p \approx 0.25$ –0.31 (Table 6)—we expand N in revision and treat RL masking as diagnostic only. (5) **Masking.** Removing *confidence* drops CR to **185.1%**/Sharpe **0.893**; removing *volatility_forecast* enlarges MDD to **−43.8%** (Table 5).

Hypothesis tests. Wilcoxon tests on daily returns versus equal-weight buy-and-hold do not reject equality at $\alpha=0.05$ for the released checkpoint; eight-seed bootstrap intervals and paired diagnostics

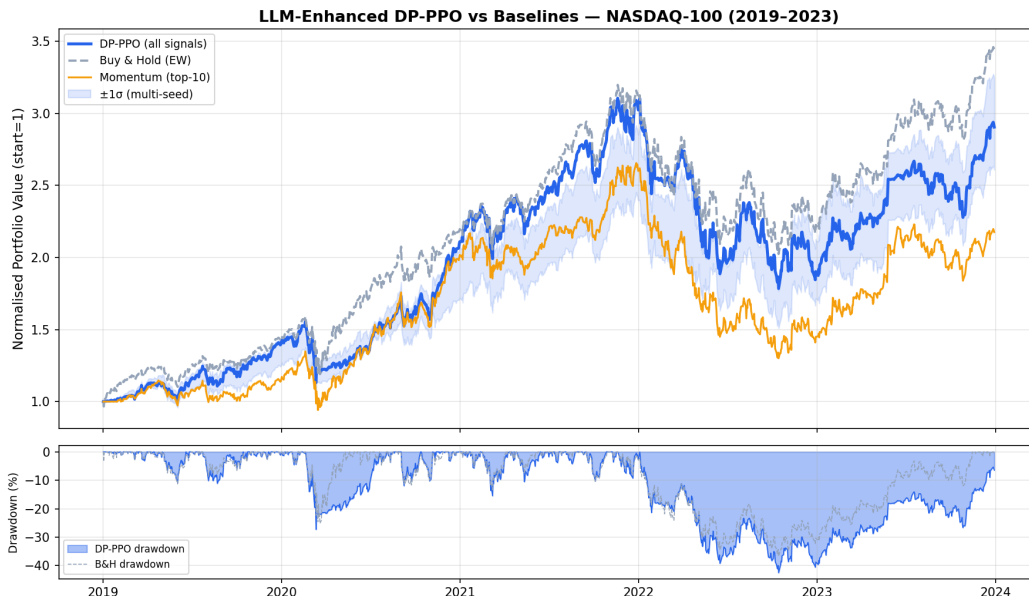


Figure 1: Normalized portfolio value on the 2019–2023 test window (DP-PPO with LLM signals versus baselines), produced by `eval_harness.py`.

in Table 9 similarly leave semantic effects statistically ambiguous at daily frequency (details in Appendix E).

5.2 Direct Semantic Factor Portfolio

Table 2 learns semantic factor weights on 2013–2018 and applies a daily long-only top-10 rule in 2019–2023. Four-factor SFP and SCW improve cumulative return and Sharpe versus sentiment-only and buy-and-hold; SRF preserves gains after residualizing non-sentiment axes on the training panel (linear diagnostic control).

Table 2: Semantic Factor Portfolio variants. Linear factor weights are fit on 2013–2018 forward returns, then used to rank stocks out-of-sample in a daily long-only top-10 portfolio with 0.1% transaction costs. SRF residualizes non-sentiment axes against sentiment. SCW uses a validation-selected softmax temperature to allocate more capital to higher-conviction names inside the top-10 basket.

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
SFP (4 factors)	307.199	1.067	1.430	-35.643	0.911
SFP (sentiment only)	291.893	1.050	1.433	-35.643	0.883
SRF (sentiment + residual axes)	307.199	1.067	1.430	-35.643	0.911
SCW (conviction-weighted)	314.075	1.067	1.460	-35.643	0.924
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.764

5.3 Supervised Forecasting Stress Test

Table 3 fits ridge models predicting 5-day returns from price/technical features with sparse LLM semantic columns, lexical headline dense features (VADER; TF-IDF/SVD), or both combined via high-conviction semantic tilt. Price-only: **126.4%** CR/Sharpe **0.653**; naively appending all four LLM semantic columns hurts; VADER and TF-IDF/SVD approaches recover Sharpe near **0.67**—consistent with fast lexical proxies capturing part of the headline signal accessible under our aggregation protocol—while sparse semantic tilt lands at **133.9%/0.674**.

Table 3: Supervised forecasting baselines. Ridge models predict 5-day forward returns from price/technical features alone, price plus LLM sentiment or four-factor semantics, price plus lexical dense headlines (VADER; train-fit TF-IDF/SVD), optional FinBERT tone when available, or a semantic tilt. Ridge strength and the semantic tilt are selected on 2018 validation Sharpe, refit on 2013–2018, and evaluated once on 2019–2023 using the same daily top-10 portfolio rule as SFP.

Strategy	Tuned	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
Supervised price-only	$\lambda = 1e + 01$	126.401	0.653	0.895	-49.963	0.356
Supervised price + sentiment	$\lambda = 1e + 00$	122.243	0.641	0.877	-49.915	0.348
Supervised price + VADER	$\lambda = 1e - 05$	134.842	0.674	0.925	-50.157	0.372
Supervised price + TF-IDF/SVD	$\lambda = 1e + 01$	132.788	0.673	0.918	-48.316	0.382
Supervised price + 4 semantic	$\lambda = 1e - 05$	111.842	0.614	0.835	-50.141	0.324
Supervised price + semantic tilt	$\lambda = 1e + 01, \alpha = 1$	133.894	0.674	0.929	-49.963	0.371
Buy & Hold (EW)	–	243.574	1.032	1.406	-36.705	0.764

5.4 Algorithm Baseline: SAC

Table 4 compares the released DP-PPO checkpoint against an SAC policy trained with the same LLM-enriched observation vector. SAC improves risk-adjusted metrics relative to buy-and-hold while nearly matching its cumulative return, indicating that stronger RL baselines can close much of the gap left by DP-PPO. This directly answers RQ3: the same sparse semantic state is more useful under an off-policy objective with replay and entropy regularisation than under the released on-policy DP-PPO run (Appendix C discusses mechanism hypotheses). We emphasise diagnostic interpretation; a controlled study isolating optimizer vs. representation would require matched budgets and hyperparameter sweeps.

Table 4: Algorithm baseline with identical four-signal observations (2019–2023). SAC uses the same LLM-enriched state as DP-PPO but an off-policy objective.

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
DP-PPO (LLM signals)	179.586	0.925	1.260	-36.582	0.625
SAC (LLM signals)	242.723	1.127	1.552	-33.168	0.844
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.764

5.5 Ablation: Signal Contribution

Table 5 reports leave-one-signal-out masking at *evaluation* time on fixed trained weights (Section 6); it is not a replacement for retraining-based causal attribution.

Table 5: Signal Ablation Study — DP-PPO Variants

Strategy	CR (%)	Sharpe	Rachev	MDD (%)
DP-PPO (all signals)	190.431	0.917	0.911	-42.540
DP-PPO (no sentiment)	191.113	0.913	0.917	-42.936
DP-PPO (no risk)	190.787	0.915	0.915	-42.705
DP-PPO (no confidence)	185.075	0.893	0.920	-43.176
DP-PPO (no volatility_forecast)	186.568	0.896	0.919	-43.780
DP-PPO (neutral)	196.754	0.926	0.921	-43.541

5.6 Multi-seed robustness

Table 6 summarises mean±std cumulative return and Sharpe across 21 seeds; *neutral eval* masks LLM coordinates at test time. Bootstrap CIs and paired Wilcoxon *p*-values are reported in the table caption and Section 5.

Table 6: Multi-seed robustness (21 seeds, 2019–2023). Mean $\pm$ std across seeds for cumulative return and Sharpe. *Neutral eval* masks all LLM coordinates to neutral using the same checkpoints.

Variant	CR (%)	Sharpe
DP-PPO (full signals)	195.04 $\pm$ 44.18	0.920 $\pm$ 0.096
DP-PPO (neutral eval)	190.93 $\pm$ 42.57	0.907 $\pm$ 0.094

Sub-period and transaction-cost sensitivity analyses are reported in Appendix E (Tables 7, 8; Figure 3). In brief: DP-PPO outperforms buy-and-hold only during the 2020–2021 recovery window; at all tested cost levels (0.05%–2.0%) DP-PPO remains below the passive sleeve.

6 Discussion

Interpretive value versus causal attribution. A single sentiment score conflates risk and opportunity. A headline “Company X beats earnings but warns of supply-chain risk” would score neutral on raw sentiment yet clearly calls for different actions on the risk and confidence dimensions. The SFP, SRF, and SCW baselines support multi-axis decomposition versus sentiment-only; coverage effects interact with the price panel rather than replacing it (Appendix F). The paired daily-return diagnostics in Table 9 show statistically ambiguous edges at daily frequency. Evaluation-time masking does *not* identify causal effects of removing channels during RL training—those require retrained policies (limitations below).

Direct versus policy-mediated use. The direct SFP/SRF/SCW baseline outperforms sentiment-only SFP and buy-and-hold (Table 2), but the released DP-PPO checkpoint trails equal-weight buy-and-hold on headline metrics while aggregate neutral masking lowers mean returns across 21 seeds (Table 6); an SAC baseline with the same LLM state nearly closes this return gap while improving risk-adjusted metrics (Table 4). The supervised forecaster in Table 3 further shows why representation matters: sparse LLM factors help the direct portfolio interface and as high-conviction overlays on price forecasts but hurt naive ridge augmentation; lexical dense headline features (VADER; TF-IDF/SVD) provide complementary supervised signals under the same portfolio rule. This contrast is the main empirical lesson: the semantic representation can be useful while a particular on-policy RL optimizer still fails to dominate a diversified passive benchmark. Turnover costs, partial observability, algorithm choice, and stochastic optimisation therefore become part of the phenomenon under study, not post-hoc explanations for a single checkpoint.

RL as diagnostic stress testing. Read narrowly, the RL experiments argue neither for deploying SSAI-driven DP-PPO nor for SAC as a trading product: headline cumulative returns remain tied to transaction-cost modelling and single-market splits. Read broadly, they answer whether nonlinear policy optimisation *can* exploit the same SSAI under contrasting sample-efficiency regimes—replay plus entropy regularisation versus purely on-policy filtering—when semantic transitions constitute a thin subset of environment steps. Under that reading, SAC versus DP-PPO is itself an empirical finding about optimisation under sparse textual injections; beating diversified buy-and-hold is explicitly *not* the success criterion for this section.

LLM signals as an engineered interface. We use a general-purpose LLM accessed through a hosted API without finance-specific fine-tuning. Structured prompts and few-shot exemplars yield stable score distributions (Table 10), and the proxy IC analysis in Table 12 shows small but interpretable alignments with realized returns and absolute returns. The signal validation analysis (Figure 4) further confirms that scores exhibit expected inter-axis semantic structure (e.g., $\rho_{\text{sentiment,risk}} = -0.756$; $\rho_{\text{risk,vol}} = +0.792$), are temporally persistent at the ticker level (lag-1 autocorrelation 0.47–0.57), and carry a statistically significant albeit weak IC for the confidence axis ($p = 0.004$). Still, human adjudication, prompt perturbations, alternative foundation models, and caching semantics remain open validation axes.

Generalizability beyond finance. The SSAI template (Definition 1) could in principle apply whenever sparse text augments sequential decisions; Appendix D lists *illustrative* cross-domain

297 sketches. **No additional domains are evaluated here**; treating those sketches as validated would
298 over-claim.

299 **Limitations and future work.** Key limitations: (1) news coverage is uneven across tickers (0–
300 27%)—performance should be stratified by coverage terciles in revision; (2) masking ablations
301 reweight fixed checkpoints and are not causal channel ablations without retraining; (3) DP-PPO
302 seed studies now use 21 seeds (non-significant Wilcoxon; overlapping bootstrap CIs for seed-mean
303 Sharpe)—further scaling and power-aware language will be reported; (4) *dense neural text* stops
304 at lexical VADER/TF-IDF/SVD (optional FinBERT column in code): a frozen FinBERT (or small
305 fine-tuned head) baseline on the *same* news window and splits is the highest-priority comparison
306 for revision; (5) LLM scoring uses a fixed prompt on a hosted endpoint—supplement will record
307 model family/version and date range; we will add a small prompt-perturbation and/or subsampled
308 human-rating study; (6) evaluation is retrospective—live slippage and impact are unmodelled. Priority
309 extensions: coverage-stratified tables, pooled-embedding baselines, prompt sensitivity, expanded
310 seeds, and rolling OOS evaluation.

311 **Ethics and reproducibility note.** This work is a retrospective research benchmark; results should
312 not be interpreted as live trading recommendations (see Appendix G for full discussion). The artifact
313 package includes scripts, CSV panels, checkpoints, and prompt templates to support reproducibility.

314 7 Conclusion

315 We present FinRL-MultiSignal, an *empirical* study of LLM semantic factorization as a state abstrac-
316 tion for US equity portfolio decisions under sparse news. The central finding: *LLM-derived semantic*
317 *signals are not effective as dense ridge features, but are effective as sparse, high-conviction allocation*
318 *signals; their utility inside RL depends critically on whether the optimisation regime can exploit rare,*
319 *high-information transitions.* Four-factor SFP/SRF/SCW outperform sentiment-only and buy-and-
320 hold in direct portfolios (307–314% CR); SAC with the same LLM state nearly matches buy-and-hold
321 (242.7%) while DP-PPO trails; eight-seed mean Sharpe 0.913 ± 0.094 (full) vs 0.889 ± 0.084 (neutral)
322 with non-significant paired tests at that N . The SSAI formalism (Appendix A), code artifact, and
323 diagnostic decomposition are reusable beyond this panel; **cross-domain empirical validation is**
324 **future work.**

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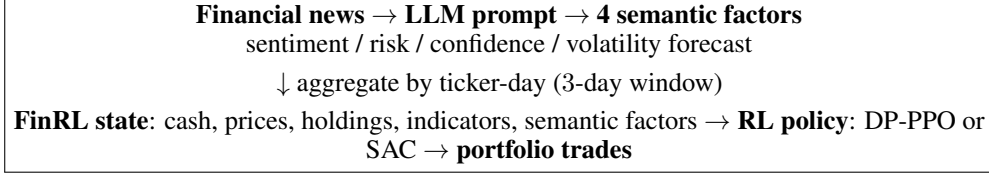


Figure 2: FinRL-MultiSignal pipeline. The LLM supplies structured semantic factors; trading decisions are learned by the RL policy from market state plus those factors.

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361 A Formal Framework Definition

362 **Definition 1 (Semantic State Abstraction Interface, SSAI).** Let $\mathcal{N}_{s,d}$ denote the set of text documents for asset s on day d . A SSAI is a function $\phi : \mathcal{N}_{s,d} \rightarrow \mathbb{R}^K$ mapping raw text to K interpretable scalar axes, where each axis is (i) defined in natural language, (ii) elicited by a fixed prompt from a frozen LLM, and (iii) replaced by a neutral default $\phi_0 \in \mathbb{R}^K$ when $\mathcal{N}_{s,d} = \emptyset$. Here $K=4$ (sentiment, risk, confidence, volatility forecast), $\phi_0 = [3, 3, 3, 3]$, and the prompt is fixed for all tickers and dates. The SSAI is *auditable* (each axis is human-interpretable), *sparse-by-design* (neutral default on no-news days), and *optimizer-agnostic* (the same ϕ feeds both direct factor rules and RL state vectors).

363 B Implementation Details

364 B.1 Pipeline schematic

365 B.2 LLM Scoring Prompt

366 The system prompt used for multi-signal scoring is:

367 You are a quantitative financial analyst. For each news snippet
368 about a stock you will output exactly four integer scores on a 1-5
369 scale, separated by a pipe '|', in this exact order: sentiment |
370 risk | confidence | volatility_forecast. (*Per-axis rubrics and*
371 *canonical few-shot exemplars are lengthy; the verbatim templates are*
372 *included with the scoring scripts in the artifact package.*) When
373 multiple news items are given for one batch, output one line per
374 item. Never add explanation -- only scores.

Two few-shot examples are prepended to each batch request. Up to 20 articles are batched per API call to amortise latency.

B.3 Network Architecture

Hyperparameter	Value
Hidden layers	2×512 (Tanh)
Observation dim	421
Action dim	30
PPO clip	0.2
Learning rate	3×10^{-4}
Discount γ	0.99
GAE λ	0.97
Steps per epoch	$4,000 \times 4$ workers
Train epochs	30
Max KL	0.01
Drawdown penalty α	0.1

B.4 Data Pipeline

Stock price data: Yahoo Finance via `yfinance` (adjusted closes, 2013-01-02–2023-12-29, 30 NASDAQ-100 tickers). Technical indicators computed via `stockstats`. Turbulence index computed following Kritzman and Li [2010]. News data: Hugging Face dataset `Zihan1004/FNSPID`, filtered to the 30 target tickers, 40,850 articles collected (2009–2023).

Ticker universe. AAPL, ADBE, ADI, AMAT, AMD, AMZN, ASML, AVGO, CDNS, COST, GOOGL, INTC, INTU, KLAC, LRCX, MCHP, META, MRVL, MSFT, MU, NFLX, NVDA, ORCL, PANW, QCOM, REGN, SNPS, TSLA, TXN, ISRG.

B.5 Compute

All experiments were run on a single workstation (Apple Silicon, 16 GB unified memory). LLM scoring: ≈ 17.5 hours for 40,850 articles at batch size 20 via a hosted chat-completions endpoint. DP-PPO training: ≈ 45 minutes per 30-epoch run (4 MPI workers, CPU-only PyTorch).

B.6 Drawdown shaping interpretation

Let $D_t = \max(0, (W_{\text{peak}} - W_t)/W_{\text{peak}})$ denote fractional drawdown. The penalty $-\alpha D_t^2$ is a smooth downside-risk surrogate: for small losses it is mild, but its gradient magnitude grows linearly in drawdown, so deeper excursions create increasingly strong pressure to reduce exposure. This differs from variance regularisation (symmetric upside/downside penalties) and from CVaR-style constraints (explicit tail optimisation). The intent is pragmatic shaping compatible with standard policy-gradient code.

C SAC mechanism hypotheses

The comparison suggests an optimisation explanation rather than a state-representation explanation alone: SAC exhibits lower realised volatility, smaller drawdown, shorter drawdown duration, and higher bear-market outperformance than DP-PPO under identical semantic inputs. A plausible mechanism is that on-policy PPO re-uses each transition only once: when only 14.8% of stock-days carry non-neutral semantic coordinates, PPO rollouts are dominated by neutral-signal steps, and rare informative transitions are weighted equally with uninformative ones. SAC’s experience replay allows semantic-rich transitions to be revisited proportionally more often during gradient updates, while entropy regularisation reduces premature policy collapse before semantic evidence accumulates. This “sparse-signal replay” effect is a property of the optimiser rather than the representation.

D Cross-domain SSAI instantiations

The SSAI framework (Definition 1) is not specific to portfolio trading *as a definition*. Any sequential system that occasionally observes sparse unstructured text *could* use a fixed-axis LLM interface as an auditable, sparse-by-design augmentation. **The present submission does not run experiments in these domains**; the following are *illustrative* sketches only: clinical triage (patient notes → urgency/confidence/risk); supply-chain optimisation (disruption news → risk/confidence/lead-time); recommendation (reviews → sentiment/engagement/quality). A diagnostic suite analogous to ours—direct factor tests, residualization, masking, optimiser comparison—*could* be instantiated elsewhere; doing so is future work. Prompts and the released harness are domain-agnostic at the software level, but all numbers in the main paper refer to equities only.

E Additional Results

E.1 Regime Analysis and Sub-period Breakdown

The 2019–2023 evaluation period spans three distinct regimes:

- **2019–2020 Q1**: bull market followed by COVID crash (S&P 500 −34% in 33 days, February–March 2020).
- **2020 Q2–2021**: recovery and growth rally.
- **2022**: Federal Reserve rate hike bear market (NASDAQ −33% in 2022).

The turbulence index exceeded the threshold of 380 on 87 trading days (6.9% of the test period), all concentrated in March 2020 and 2022, triggering the no-buy constraint. Table 7 reports a more granular sub-period breakdown for the released checkpoint. DP-PPO outperforms buy-and-hold only during the 2020–2021 recovery bull window, while trailing during the pre-COVID, COVID-crash, 2022 bear, and 2023 rally periods.

Table 7: Sub-period breakdown for the released DP-PPO checkpoint versus equal-weight buy-and-hold.

Period	Days	DP-PPO CR	B&H CR	Δ CR	Sharpe
Pre-COVID (2019–Feb 2020)	292	34.01	41.80	-7.79	1.400
COVID Crash (Mar–Apr 2020)	43	-9.80	2.45	-12.25	-1.145
Recovery Bull (May 2020–Dec 2021)	422	144.86	110.56	34.30	2.159
Rate Hike Bear (2022)	251	-38.91	-28.99	-9.92	-1.194
2023 Rally (2023)	250	54.91	55.44	-0.53	2.163

E.2 Transaction-Cost Sensitivity

Table 8 sweeps per-trade costs from 0.05% to 2.0% for the released DP-PPO checkpoint. Because buy-and-hold incurs only the initial allocation in this simplified sweep, DP-PPO remains below the passive sleeve at every tested cost level.

Table 8: Transaction-cost sensitivity for the released DP-PPO checkpoint. Buy-and-hold has no ongoing rebalance cost in this sweep; DP-PPO remains below the passive sleeve across tested cost levels.

Cost (%)	DP-PPO CR	B&H CR	Δ CR	Sharpe	MDD (%)
0.05	188.07	243.57	-55.51	0.908	-42.81
0.10	190.43	243.57	-53.14	0.917	-42.54
0.20	185.72	243.57	-57.86	0.899	-43.49
1.00	165.65	243.57	-77.93	0.840	-44.85
2.00	141.43	243.57	-102.14	0.769	-45.57

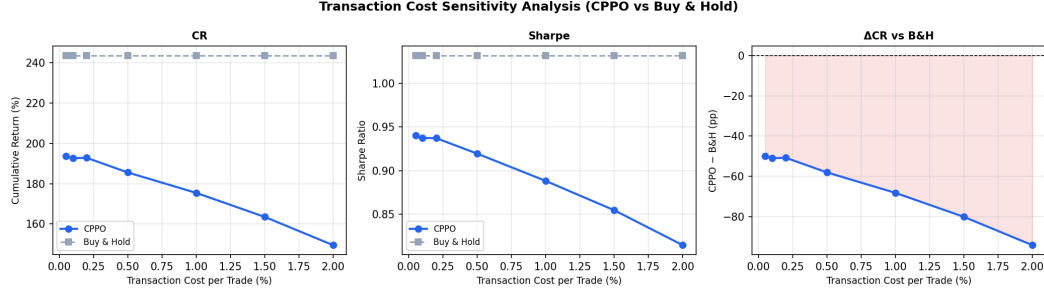


Figure 3: Transaction-cost sensitivity for the released DP-PPO checkpoint versus equal-weight buy-and-hold.

E.3 Paired daily-return diagnostics (non-RL baselines)

Table 9: Paired daily-return diagnostics: semantic baselines plus multi-seed RL vs EW benchmark. The confidence interval is a 20-trading-day block bootstrap over mean paired daily active returns, in basis points per day. Wilcoxon tests are paired over daily returns and are reported as diagnostics rather than definitive multiple-comparison-adjusted claims.

Comparison	Mean bp/day	95% CI	Δ Sharpe	Win %	Wilcoxon p
SFP 4-factor vs sentiment-only	0.352	[-1.070, 1.885]	+0.017	27.4	0.556
SCW vs equal-weight SFP	0.182	[-1.224, 1.740]	+0.000	24.8	0.232
Semantic tilt vs price-only forecaster	0.234	[-1.198, 1.497]	+0.021	34.8	0.568
Multi-seed DP-PPO mean vs EW buy-and-hold	-1.392	[-4.119, 1.606]	-0.082	49.7	0.631

F LLM Signal Validation

Table 10: Descriptive statistics of LLM-generated signal scores (39,995 articles).

	Sentiment	Risk	Confidence	Vol. Forecast
Mean	3.35	2.47	3.51	2.74
Std	1.01	1.02	0.77	0.82
Median	3.00	2.00	3.00	3.00

G Ethics, Deployment, and Reproducibility

This work is a retrospective research benchmark, not a trading recommendation or deployable advisory system. Automated trading systems can amplify losses, liquidity stress, and feedback loops if deployed without governance, risk limits, and human oversight. Our evaluation omits important operational frictions—slippage, partial fills, exchange outages, short-sale constraints, tax effects, and capital limits—so reported results should not be interpreted as expected live performance. The LLM component introduces additional reproducibility concerns: hosted model behaviour can drift over time; prompts may elicit different outputs across providers or versions; and financial news coverage is uneven across firms. Scored article-level signals should be cached and released with prompt templates, model identifiers, timestamps, and dataset hashes. The artifact package provides scripts, processed CSV panels, checkpoints, and dry-run commands for retrained ablations, but full reproducibility still depends on access to the same market data snapshot and LLM-scored signal cache.

LLM Semantic Signal Validation

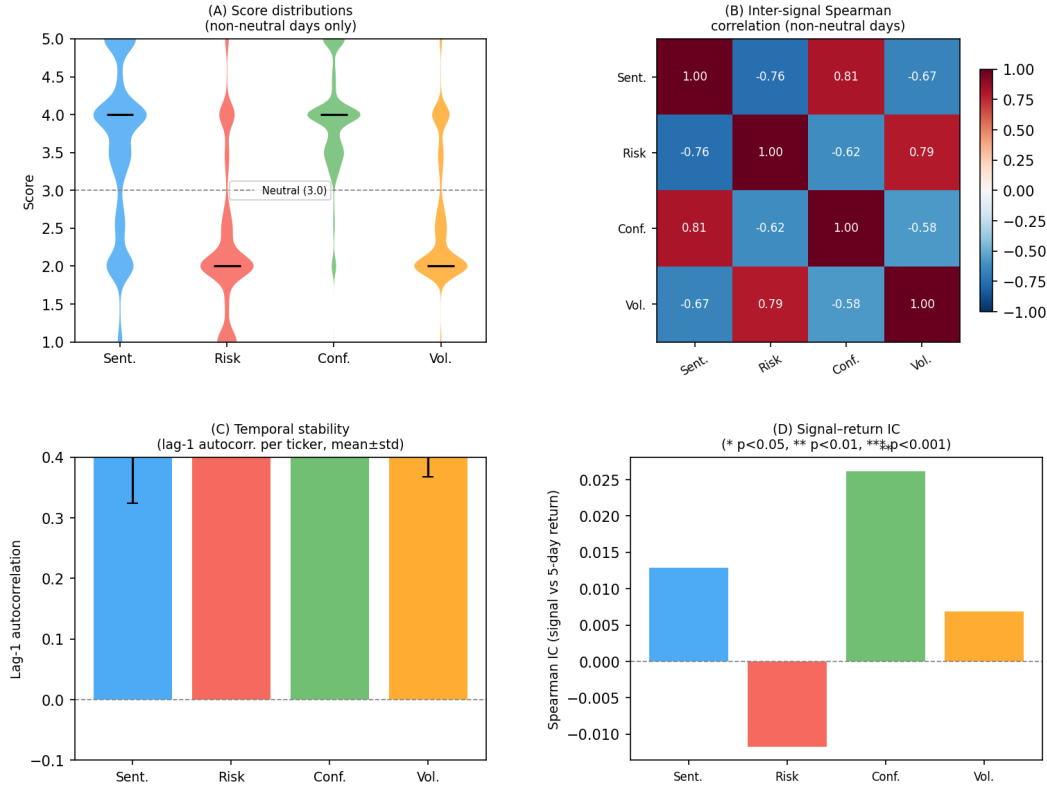


Figure 4: LLM semantic signal validation on the full 2013–2023 panel ($N=83,040$ stock-days, 18,578 non-neutral). (A) Score distributions on non-neutral days show structured directional skew. (B) Inter-signal Spearman correlations replicate expected semantic structure. (C) Lag-1 temporal autocorrelation per ticker confirms scores are not white noise. (D) Signal–return IC: confidence shows a significant positive IC ($p=0.004$).

Table 11: Ticker-level LLM signal coverage on the 2019–2023 trade panel. Coverage is the percentage of stock-days whose score differs from the neutral imputation value 3.0; rows show the five most-covered and five least-covered tickers.

Ticker	Stock-days	Any (%)	Sent.	Risk	Conf.	Vol.
GOOGL	1258	26.9	22.1	23.4	19.0	22.4
AVGO	1258	26.6	19.8	24.4	16.6	23.1
NVDA	1258	26.5	22.1	23.4	19.0	20.5
MU	1258	24.6	21.2	22.8	18.0	21.4
ASML	1258	23.4	17.9	19.5	13.3	16.1
INTC	1258	0.5	0.5	0.5	0.5	0.5
META	1258	0.0	0.0	0.0	0.0	0.0
MSFT	1258	0.0	0.0	0.0	0.0	0.0
ISRG	1258	0.0	0.0	0.0	0.0	0.0
COST	1258	0.0	0.0	0.0	0.0	0.0

Table 12: Signal validity proxy: pooled Spearman information coefficients between LLM signal coordinates and 5-day forward returns / absolute returns on the 2019–2023 stock-date panel. This is not a human annotation audit, but it checks whether scores align with realized market quantities rather than pure noise.

Signal	Return IC	p	Return IC	p
sentiment	0.0044	0.39	-0.0047	0.36
risk	-0.0152	0.0032	0.0122	0.018
confidence	0.0217	2.5e-05	0.0065	0.21
volatility_forecast	-0.0045	0.38	0.0175	0.00068

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